

Uncertainty principle

The **uncertainty principle**, also known as **Heisenberg's indeterminacy principle**, is a fundamental concept in [quantum mechanics](#). It states that there is a limit to the precision with which certain pairs of physical properties, such as position and [momentum](#), can be simultaneously known. In other words, the more accurately one property is measured, the less accurately the other property can be known.

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W. Heisenberg,

ermöglichen, als es der Gleichung (1) entspricht, so wäre die Quantenmechanik unmöglich. Diese Ungenauigkeit, die durch Gleichung (1) festgelegt ist, schafft also erst Raum für die Gültigkeit der Beziehungen, die in den quantenmechanischen Vertauschungsrelationen

$$pq - qp = \frac{\hbar}{2\pi i}$$

ihren prägnanten Ausdruck finden; sie ermöglicht diese Gleichung, ohne daß der physikalische Sinn der Größen p und q geändert werden mußte.

Canonical commutation rule for position q and momentum p variables of a [particle](#), 1927. $pq - qp = h/(2\pi i)$. Uncertainty principle of Heisenberg, 1927.



German physicist [Werner Heisenberg](#) (1901–1976), who introduced Uncertainty Principle in 1927

More formally, the uncertainty principle is any of a variety of [mathematical inequalities](#) asserting a fundamental limit to the product of the accuracy of certain related pairs of measurements on a

quantum system, such as [position](#), x , and momentum, p .^[1] Such paired-variables are known as [complementary variables](#) or [canonically conjugate variables](#).

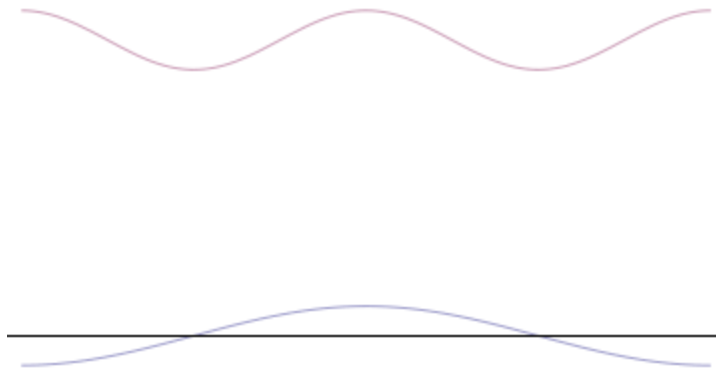
First introduced in 1927 by German physicist [Werner Heisenberg](#),^{[2][3][4][5]} the formal inequality relating the [standard deviation](#) of position σ_x and the standard deviation of momentum σ_p was derived by [Earle Hesse Kennard](#)^[6] later that year and by [Hermann Weyl](#)^[7] in 1928:

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}$$

where $\hbar = \frac{h}{2\pi}$ is the [reduced Planck constant](#).

The quintessentially quantum mechanical uncertainty principle comes in many forms other than position–momentum. The energy–time relationship is widely used to relate quantum state lifetime to measured energy widths but its formal derivation is fraught with confusing issues about the nature of time. The basic principle has been extended in numerous directions; it must be considered in many kinds of fundamental physical measurements.

Position–momentum



The superposition of several plane waves to form a wave packet. This wave packet becomes increasingly localized with the addition of many waves. The Fourier transform is a mathematical operation that separates a wave packet into its individual plane waves. The waves shown here are [real](#) for illustrative purposes only; in quantum mechanics the wave function is generally [complex](#).

It is vital to illustrate how the principle applies to relatively intelligible physical situations since it is indiscernible on the macroscopic^[8] scales that humans experience. Two alternative frameworks for

quantum physics offer different explanations for the uncertainty principle. The [wave mechanics](#) picture of the uncertainty principle is more visually intuitive, but the more abstract [matrix mechanics](#) picture formulates it in a way that generalizes more easily. The effect of Heisenberg uncertainty principle is significant only for motion of microscopic objects and is negligible for that of macroscopic objects .

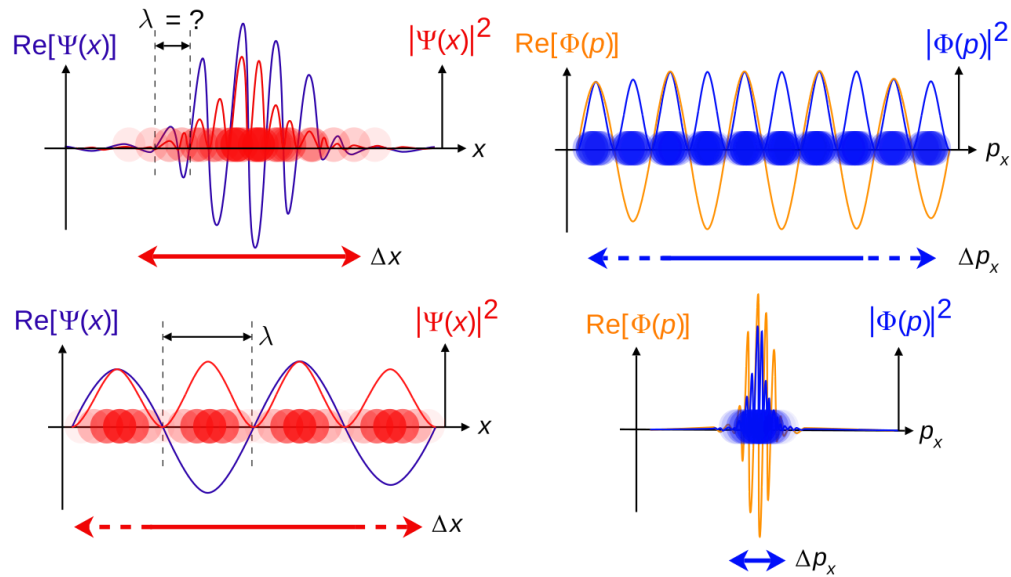
Mathematically, in wave mechanics, the uncertainty relation between position and momentum arises because the expressions of the wavefunction in the two corresponding [orthonormal bases](#) in [Hilbert space](#) are [Fourier transforms](#) of one another (i.e., position and momentum are [conjugate variables](#)). A nonzero function and its Fourier transform cannot both be sharply localized at the same time.^[9] A similar tradeoff between the variances of Fourier conjugates arises in all systems underlain by Fourier analysis, for example in sound waves: A pure tone is a [sharp spike](#) at a single frequency, while its Fourier transform gives the shape of the sound wave in the time domain, which is a completely delocalized sine wave. In quantum mechanics, the two key points are that the position of the particle takes the form of a matter wave, and momentum is its Fourier conjugate, assured by the [de Broglie relation](#) $p = \hbar k$, where k is the [wavenumber](#).

In [matrix mechanics](#), the [mathematical formulation of quantum mechanics](#), any pair of non-[commuting self-adjoint operators](#) representing [observables](#) are subject to similar uncertainty limits. An eigenstate of an observable represents the state of the wavefunction for a certain measurement value (the eigenvalue). For example, if a measurement of an observable A is performed, then the system is in a particular eigenstate Ψ of that observable. However, the particular eigenstate of the observable A need not be an eigenstate of another observable B : If so, then it does not have a unique associated measurement for it, as the system is not in an eigenstate of that observable.^[10]

Visualization

The uncertainty principle can be visualized using the position- and momentum-space wavefunctions for one spinless [particle](#) with mass in one dimension.

The more localized the position-space wavefunction, the more likely the particle is to be found with the position coordinates in that region, and correspondingly the momentum-space wavefunction is less localized so the possible momentum components the particle could have are more widespread. Conversely, the more localized the momentum-space wavefunction, the more likely the particle is to be found with those values of momentum components in that region, and correspondingly the less localized the position-space wavefunction, so the position coordinates the particle could occupy are more widespread. These wavefunctions are [Fourier transforms](#) of each other: mathematically, the uncertainty principle expresses the relationship between conjugate variables in the transform.



Position x and momentum p wavefunctions corresponding to quantum particles. The colour opacity of the particles corresponds to the [probability density](#) of finding the particle with position x or momentum component p .

Top: If wavelength λ is unknown, so are momentum p , wave-vector k and energy E (de Broglie relations). As the particle is more localized in position space, Δx is smaller than for Δp_x .

Bottom: If λ is known, so are p , k , and E . As the particle is more localized in momentum space, Δp is smaller than for Δx .

Wave mechanics interpretation

According to the [de Broglie hypothesis](#), every object in the universe is associated with a [wave](#). Thus every object, from an elementary particle to atoms, molecules and on up to planets and beyond are subject to the uncertainty principle.

The time-independent wave function of a single-moded plane wave of wavenumber k_0 or momentum p_0 is^[11]

$$\psi(x) \propto e^{ik_0x} = e^{ip_0x/\hbar}.$$

The [Born rule](#) states that this should be interpreted as a [probability density amplitude function](#) in the sense that the probability of finding the particle between a and b is

$$P[a \leq X \leq b] = \int_a^b |\psi(x)|^2 dx.$$

In the case of the single-mode plane wave, $|\psi(x)|^2$ is 1 if $X = x$ and 0 otherwise. In other words, the particle position is extremely uncertain in the sense that it could be essentially anywhere along the wave packet.

On the other hand, consider a wave function that is a [sum of many waves](#), which we may write as

$$\psi(x) \propto \sum_n A_n e^{ip_n x / \hbar},$$

where A_n represents the relative contribution of the mode p_n to the overall total. The figures to the right show how with the addition of many plane waves, the wave packet can become more localized. We may take this a step further to the [continuum limit](#), where the wave function is an [integral](#) over all possible modes

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \varphi(p) \cdot e^{ipx/\hbar} dp,$$

with $\varphi(p)$ representing the amplitude of these modes and is called the wave function in [momentum space](#). In mathematical terms, we say that $\varphi(p)$ is the [Fourier transform](#) of $\psi(x)$ and that x and p are [conjugate variables](#). Adding together all of these plane waves comes at a cost, namely the momentum has become less precise, having become a mixture of waves of many different momenta.^[12]

One way to quantify the precision of the position and momentum is the [standard deviation](#) σ . Since $|\psi(x)|^2$ is a probability density function for position, we calculate its standard deviation.

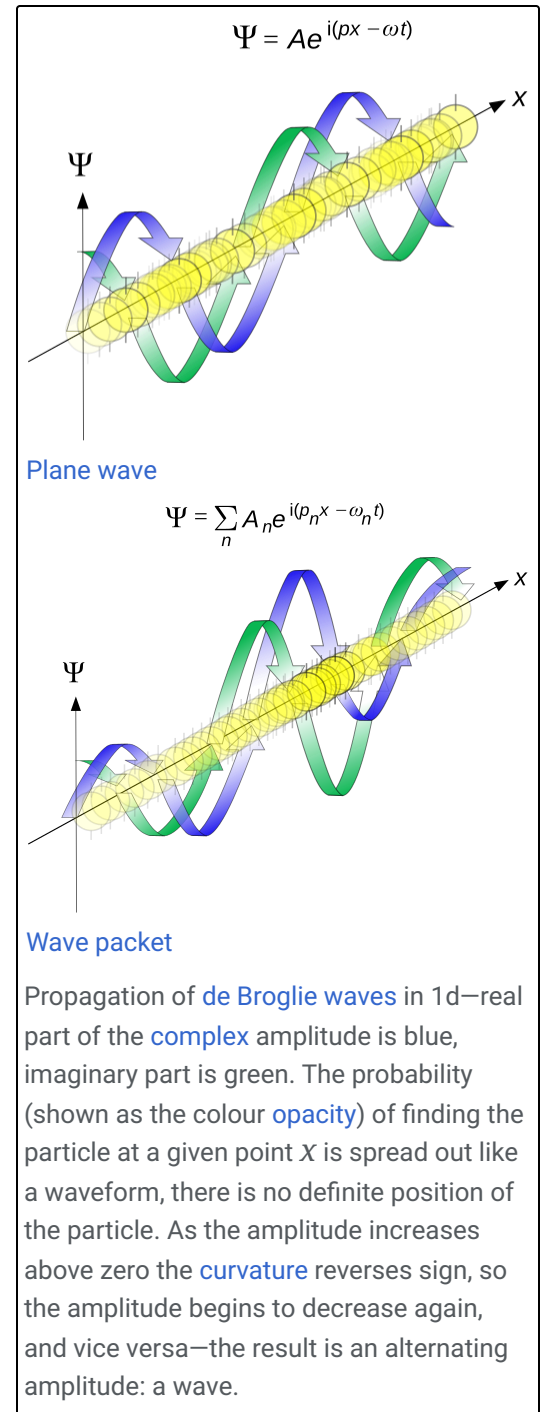
The precision of the position is improved, i.e. reduced σ_x , by using many plane waves, thereby weakening the precision of the momentum, i.e. increased σ_p . Another way of stating this is that σ_x and σ_p have an [inverse relationship](#) or are at least bounded from below. This is the uncertainty principle, the exact limit of which is the Kennard bound.

Proof of the Kennard inequality using wave mechanics

We are interested in the [variances](#) of position and momentum, defined as

$$\sigma_x^2 = \int_{-\infty}^{\infty} x^2 \cdot |\psi(x)|^2 dx - \left(\int_{-\infty}^{\infty} x \cdot |\psi(x)|^2 dx \right)^2$$

$$\sigma_p^2 = \int_{-\infty}^{\infty} p^2 \cdot |\varphi(p)|^2 dp - \left(\int_{-\infty}^{\infty} p \cdot |\varphi(p)|^2 dp \right)^2.$$



Without loss of generality, we will assume that the means vanish, which just amounts to a shift of the origin of our coordinates. (A more general proof that does not make this assumption is given below.) This gives us the simpler form

$$\sigma_x^2 = \int_{-\infty}^{\infty} x^2 \cdot |\psi(x)|^2 dx$$

$$\sigma_p^2 = \int_{-\infty}^{\infty} p^2 \cdot |\varphi(p)|^2 dp .$$

The function $f(x) = x \cdot \psi(x)$ can be interpreted as a vector in a function space. We can define an inner product for a pair of functions $u(x)$ and $v(x)$ in this vector space:

$$\langle u | v \rangle = \int_{-\infty}^{\infty} u^*(x) \cdot v(x) dx,$$

where the asterisk denotes the complex conjugate.

With this inner product defined, we note that the variance for position can be written as

$$\sigma_x^2 = \int_{-\infty}^{\infty} |f(x)|^2 dx = \langle f | f \rangle .$$

We can repeat this for momentum by interpreting the function $\tilde{g}(p) = p \cdot \varphi(p)$ as a vector, but we can also take advantage of the fact that $\psi(x)$ and $\varphi(p)$ are Fourier transforms of each other. We evaluate the inverse Fourier transform through integration by parts:

$$\begin{aligned}
g(x) &= \frac{1}{\sqrt{2\pi\hbar}} \cdot \int_{-\infty}^{\infty} \tilde{g}(p) \cdot e^{ipx/\hbar} dp \\
&= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} p \cdot \varphi(p) \cdot e^{ipx/\hbar} dp \\
&= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \left[p \cdot \int_{-\infty}^{\infty} \psi(\chi) e^{-ip\chi/\hbar} d\chi \right] \cdot e^{ipx/\hbar} dp \\
&= \frac{i}{2\pi} \int_{-\infty}^{\infty} \left[\cancel{\psi(\chi) e^{-ip\chi/\hbar}} \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{d\psi(\chi)}{d\chi} e^{-ip\chi/\hbar} d\chi \right] \cdot e^{ipx/\hbar} dp \\
&= -i \int_{-\infty}^{\infty} \frac{d\psi(\chi)}{d\chi} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ip(x-\chi)/\hbar} dp \right] d\chi \\
&= -i \int_{-\infty}^{\infty} \frac{d\psi(\chi)}{d\chi} \left[\delta\left(\frac{x-\chi}{\hbar}\right) \right] d\chi \\
&= -i\hbar \int_{-\infty}^{\infty} \frac{d\psi(\chi)}{d\chi} [\delta(x-\chi)] d\chi \\
&= -i\hbar \frac{d\psi(x)}{dx} \\
&= \left(-i\hbar \frac{d}{dx} \right) \cdot \psi(x),
\end{aligned}$$

where $v = \frac{\hbar}{-ip} e^{-ip\chi/\hbar}$ in the integration by parts, the cancelled term vanishes because the wave function vanishes at both infinities and $|e^{-ip\chi/\hbar}| = 1$, and then use the [Dirac delta function](#) which is valid because $\frac{d\psi(\chi)}{d\chi}$ does not depend on p .

The term $-i\hbar \frac{d}{dx}$ is called the [momentum operator](#) in position space. Applying [Plancherel's theorem](#), we see that the variance for momentum can be written as

$$\sigma_p^2 = \int_{-\infty}^{\infty} |\tilde{g}(p)|^2 dp = \int_{-\infty}^{\infty} |g(x)|^2 dx = \langle g | g \rangle.$$

The [Cauchy-Schwarz inequality](#) asserts that

$$\sigma_x^2 \sigma_p^2 = \langle f | f \rangle \cdot \langle g | g \rangle \geq |\langle f | g \rangle|^2.$$

The [modulus squared](#) of any complex number z can be expressed as

$$|z|^2 = \left(\text{Re}(z) \right)^2 + \left(\text{Im}(z) \right)^2 \geq \left(\text{Im}(z) \right)^2 = \left(\frac{z - z^*}{2i} \right)^2.$$

we let $z = \langle f | g \rangle$ and $z^* = \langle g | f \rangle$ and substitute these into the equation above to get

$$|\langle f | g \rangle|^2 \geq \left(\frac{\langle f | g \rangle - \langle g | f \rangle}{2i} \right)^2.$$

All that remains is to evaluate these inner products.

$$\begin{aligned} \langle f | g \rangle - \langle g | f \rangle &= \int_{-\infty}^{\infty} \psi^*(x) x \cdot \left(-i\hbar \frac{d}{dx} \right) \psi(x) dx - \int_{-\infty}^{\infty} \psi^*(x) \left(-i\hbar \frac{d}{dx} \right) \cdot x \psi(x) dx \\ &= i\hbar \cdot \int_{-\infty}^{\infty} \psi^*(x) \left[\left(-x \cdot \frac{d\psi(x)}{dx} \right) + \frac{d(x\psi(x))}{dx} \right] dx \\ &= i\hbar \cdot \int_{-\infty}^{\infty} \psi^*(x) \left[\left(-x \cdot \frac{d\psi(x)}{dx} \right) + \psi(x) + \left(x \cdot \frac{d\psi(x)}{dx} \right) \right] dx \\ &= i\hbar \cdot \int_{-\infty}^{\infty} \psi^*(x) \psi(x) dx \\ &= i\hbar \cdot \int_{-\infty}^{\infty} |\psi(x)|^2 dx \\ &= i\hbar \end{aligned}$$

Plugging this into the above inequalities, we get

$$\sigma_x^2 \sigma_p^2 \geq |\langle f | g \rangle|^2 \geq \left(\frac{\langle f | g \rangle - \langle g | f \rangle}{2i} \right)^2 = \left(\frac{i\hbar}{2i} \right)^2 = \frac{\hbar^2}{4}$$

and taking the square root

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}.$$

with equality if and only if p and x are linearly dependent. Note that the only *physics* involved in this proof was that $\psi(x)$ and $\varphi(p)$ are wave functions for position and momentum, which are Fourier transforms of each other. A similar result would hold for *any* pair of conjugate variables.

Matrix mechanics interpretation

In matrix mechanics, observables such as position and momentum are represented by self-adjoint operators.^[12] When considering pairs of observables, an important quantity is the *commutator*. For a pair of operators $\hat{A}$ and $\hat{B}$, one defines their commutator as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}.$$

In the case of position and momentum, the commutator is the [canonical commutation relation](#)

$$[\hat{x}, \hat{p}] = i\hbar.$$

The physical meaning of the non-commutativity can be understood by considering the effect of the commutator on position and momentum [eigenstates](#). Let $|\psi\rangle$ be a right eigenstate of position with a constant eigenvalue x_0 . By definition, this means that $\hat{x}|\psi\rangle = x_0|\psi\rangle$. Applying the commutator to $|\psi\rangle$ yields

$$[\hat{x}, \hat{p}]|\psi\rangle = (\hat{x}\hat{p} - \hat{p}\hat{x})|\psi\rangle = (\hat{x} - x_0\hat{I})\hat{p}|\psi\rangle = i\hbar|\psi\rangle,$$

where $\hat{I}$ is the [identity operator](#).

Suppose, for the sake of [proof by contradiction](#), that $|\psi\rangle$ is also a right eigenstate of momentum, with constant eigenvalue p_0 . If this were true, then one could write

$$(\hat{x} - x_0\hat{I})\hat{p}|\psi\rangle = (\hat{x} - x_0\hat{I})p_0|\psi\rangle = (x_0\hat{I} - x_0\hat{I})p_0|\psi\rangle = 0.$$

On the other hand, the above canonical commutation relation requires that

$$[\hat{x}, \hat{p}]|\psi\rangle = i\hbar|\psi\rangle \neq 0.$$

This implies that no quantum state can simultaneously be both a position and a momentum eigenstate.

When a state is measured, it is projected onto an eigenstate in the basis of the relevant observable. For example, if a particle's position is measured, then the state amounts to a position eigenstate. This means that the state is *not* a momentum eigenstate, however, but rather it can be represented as a sum of multiple momentum basis eigenstates. In other words, the momentum must be less precise. This precision may be quantified by the standard deviations,

$$\sigma_x = \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2}$$

$$\sigma_p = \sqrt{\langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2}.$$

As in the wave mechanics interpretation above, one sees a tradeoff between the respective precisions of the two, quantified by the uncertainty principle.

Quantum harmonic oscillator stationary states

Consider a one-dimensional [quantum harmonic oscillator](#). It is possible to express the position and momentum operators in terms of the [creation and annihilation operators](#):

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger)$$

$$\hat{p} = i\sqrt{\frac{m\omega\hbar}{2}}(a^\dagger - a).$$

Using the standard rules for creation and annihilation operators on the energy eigenstates,

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

$$a|n\rangle = \sqrt{n}|n-1\rangle,$$

the variances may be computed directly,

$$\sigma_x^2 = \frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right)$$

$$\sigma_p^2 = \hbar m\omega \left(n + \frac{1}{2} \right).$$

The product of these standard deviations is then

$$\sigma_x \sigma_p = \hbar \left(n + \frac{1}{2} \right) \geq \frac{\hbar}{2}.$$

In particular, the above Kennard bound^[6] is saturated for the [ground state](#) $n=0$, for which the probability density is just the [normal distribution](#).

Quantum harmonic oscillators with Gaussian initial condition

In a quantum harmonic oscillator of characteristic angular frequency ω , place a state that is offset from the bottom of the potential by some displacement x_0 as

$$\psi(x) = \left(\frac{m\Omega}{\pi\hbar} \right)^{1/4} \exp \left(-\frac{m\Omega(x - x_0)^2}{2\hbar} \right),$$

where Ω describes the width of the initial state but need not be the same as ω . Through integration over the [propagator](#), we can solve for the full time-dependent solution. After many cancelations, the probability densities reduce to

$$|\Psi(x, t)|^2 \sim \mathcal{N} \left(x_0 \cos(\omega t), \frac{\hbar}{2m\Omega} \left(\cos^2(\omega t) + \frac{\Omega^2}{\omega^2} \sin^2(\omega t) \right) \right)$$

$$|\Phi(p, t)|^2 \sim \mathcal{N}\left(-mx_0\omega \sin(\omega t), \frac{\hbar m \Omega}{2} \left(\cos^2(\omega t) + \frac{\omega^2}{\Omega^2} \sin^2(\omega t)\right)\right)$$

where we have used the notation $\mathcal{N}(\mu, \sigma^2)$ to denote a normal distribution of mean μ and variance σ^2 . Copying the variances above and applying [trigonometric identities](#), we can write the product of the standard deviations as

$$\begin{aligned} \sigma_x \sigma_p &= \frac{\hbar}{2} \sqrt{\left(\cos^2(\omega t) + \frac{\Omega^2}{\omega^2} \sin^2(\omega t)\right) \left(\cos^2(\omega t) + \frac{\omega^2}{\Omega^2} \sin^2(\omega t)\right)} \\ &= \frac{\hbar}{4} \sqrt{3 + \frac{1}{2} \left(\frac{\Omega^2}{\omega^2} + \frac{\omega^2}{\Omega^2}\right) - \left(\frac{1}{2} \left(\frac{\Omega^2}{\omega^2} + \frac{\omega^2}{\Omega^2}\right) - 1\right) \cos(4\omega t)} \end{aligned}$$

From the relations

$$\frac{\Omega^2}{\omega^2} + \frac{\omega^2}{\Omega^2} \geq 2, \quad |\cos(4\omega t)| \leq 1,$$

we can conclude the following (the right most equality holds only when $\Omega = \omega$):

$$\sigma_x \sigma_p \geq \frac{\hbar}{4} \sqrt{3 + \frac{1}{2} \left(\frac{\Omega^2}{\omega^2} + \frac{\omega^2}{\Omega^2}\right) - \left(\frac{1}{2} \left(\frac{\Omega^2}{\omega^2} + \frac{\omega^2}{\Omega^2}\right) - 1\right)}$$

Coherent states

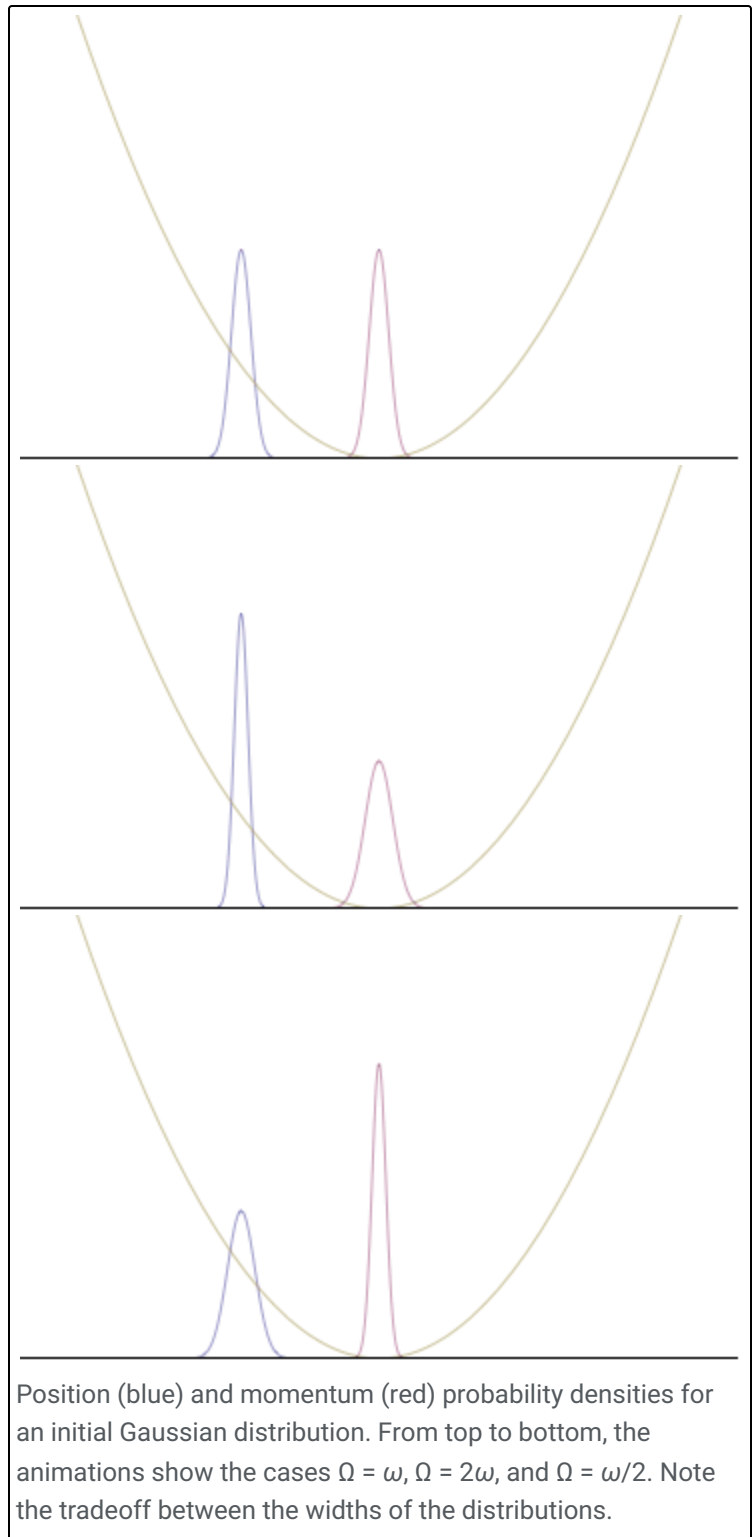
A coherent state is a right eigenstate of the [annihilation operator](#),

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle,$$

which may be represented in terms of [Fock states](#) as

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

In the picture where the coherent state is a massive particle in a quantum harmonic oscillator, the position and momentum operators may be expressed in terms of the annihilation operators in the same formulas above and used to calculate the variances,



$$\sigma_x^2 = \frac{\hbar}{2m\omega},$$

$$\sigma_p^2 = \frac{\hbar m\omega}{2}.$$

Therefore, every coherent state saturates the Kennard bound

$$\sigma_x \sigma_p = \sqrt{\frac{\hbar}{2m\omega}} \sqrt{\frac{\hbar m\omega}{2}} = \frac{\hbar}{2}.$$

with position and momentum each contributing an amount $\sqrt{\hbar/2}$ in a "balanced" way. Moreover, every [squeezed coherent state](#) also saturates the Kennard bound although the individual contributions of position and momentum need not be balanced in general.

Particle in a box

Consider a particle in a one-dimensional box of length L . The [eigenfunctions in position and momentum space](#) are

$$\psi_n(x, t) = \begin{cases} A \sin(k_n x) e^{-i\omega_n t}, & 0 < x < L, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$\varphi_n(p, t) = \sqrt{\frac{\pi L}{\hbar}} \frac{n (1 - (-1)^n e^{-ikL}) e^{-i\omega_n t}}{\pi^2 n^2 - k^2 L^2},$$

where $\omega_n = \frac{\pi^2 \hbar n^2}{8L^2 m}$ and we have used the [de Broglie relation](#) $p = \hbar k$. The variances of x and p can be calculated explicitly:

$$\sigma_x^2 = \frac{L^2}{12} \left(1 - \frac{6}{n^2 \pi^2} \right)$$

$$\sigma_p^2 = \left(\frac{\hbar n \pi}{L} \right)^2.$$

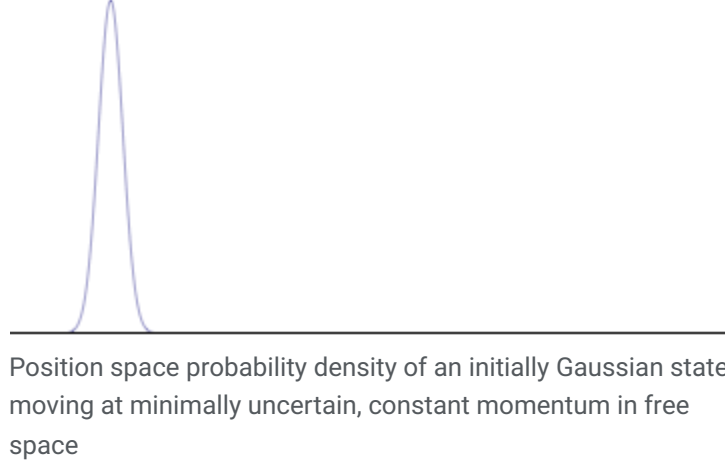
The product of the standard deviations is therefore

$$\sigma_x \sigma_p = \frac{\hbar}{2} \sqrt{\frac{n^2 \pi^2}{3} - 2}.$$

For all $n = 1, 2, 3, \dots$, the quantity $\sqrt{\frac{n^2 \pi^2}{3} - 2}$ is greater than 1, so the uncertainty principle is never violated. For numerical concreteness, the smallest value occurs when $n = 1$, in which case

$$\sigma_x \sigma_p = \frac{\hbar}{2} \sqrt{\frac{\pi^2}{3} - 2} \approx 0.568\hbar > \frac{\hbar}{2}.$$

Constant momentum



Assume a particle initially has a [momentum space](#) wave function described by a normal distribution around some constant momentum p_0 according to

$$\varphi(p) = \left(\frac{x_0}{\hbar\sqrt{\pi}} \right)^{1/2} \exp\left(\frac{-x_0^2(p - p_0)^2}{2\hbar^2} \right),$$

where we have introduced a reference scale $x_0 = \sqrt{\hbar/m\omega_0}$, with $\omega_0 > 0$ describing the width of the distribution—cf. [nondimensionalization](#). If the state is allowed to evolve in free space, then the time-dependent momentum and position space wave functions are

$$\Phi(p, t) = \left(\frac{x_0}{\hbar\sqrt{\pi}} \right)^{1/2} \exp\left(\frac{-x_0^2(p - p_0)^2}{2\hbar^2} - \frac{ip^2 t}{2m\hbar} \right),$$

$$\Psi(x, t) = \left(\frac{1}{x_0\sqrt{\pi}} \right)^{1/2} \frac{e^{-x_0^2 p_0^2 / 2\hbar^2}}{\sqrt{1 + i\omega_0 t}} \exp\left(-\frac{(x - ix_0^2 p_0 / \hbar)^2}{2x_0^2(1 + i\omega_0 t)} \right).$$

Since $\langle p(t) \rangle = p_0$ and $\sigma_p(t) = \hbar/(\sqrt{2}x_0)$, this can be interpreted as a particle moving along with constant momentum at arbitrarily high precision. On the other hand, the standard deviation of the position is

$$\sigma_x = \frac{x_0}{\sqrt{2}} \sqrt{1 + \omega_0^2 t^2}$$

such that the uncertainty product can only increase with time as

$$\sigma_x(t)\sigma_p(t) = \frac{\hbar}{2} \sqrt{1 + \omega_0^2 t^2}$$

Mathematical formalism

Starting with Kennard's derivation of position-momentum uncertainty, [Howard Percy Robertson](#) developed^{[13][1]} a formulation for arbitrary [Hermitian operators](#) $\hat{O}$ expressed in terms of their standard deviation

$$\sigma_O = \sqrt{\langle \hat{O}^2 \rangle - \langle \hat{O} \rangle^2},$$

where the brackets $\langle \hat{O} \rangle$ indicate an [expectation value](#) of the observable represented by operator $\hat{O}$. For a pair of operators $\hat{A}$ and $\hat{B}$, define their commutator as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A},$$

and the Robertson uncertainty relation is given by^[11]

$$\sigma_A \sigma_B \geq \left| \frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right| = \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right|.$$

[Erwin Schrödinger](#)^[14] showed how to allow for correlation between the operators, giving a stronger inequality, known as the **Robertson–Schrödinger uncertainty relation**,^{[15][1]}

$$\sigma_A^2 \sigma_B^2 \geq \left| \frac{1}{2} \langle \{\hat{A}, \hat{B}\} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle \right|^2 + \left| \frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right|^2,$$

where the anticommutator, $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$ is used.

Proof of the [Schrödinger](#) uncertainty relation

The derivation shown here incorporates and builds off of those shown in Robertson,^[13] Schrödinger^[15] and standard textbooks such as Griffiths.^{[16]:138} For any Hermitian operator $\hat{A}$, based upon the definition of [variance](#), we have

$$\sigma_A^2 = \langle (\hat{A} - \langle \hat{A} \rangle) \Psi | (\hat{A} - \langle \hat{A} \rangle) \Psi \rangle.$$

we let $|f\rangle = |(\hat{A} - \langle \hat{A} \rangle) \Psi\rangle$ and thus

$$\sigma_A^2 = \langle f | f \rangle.$$

Similarly, for any other Hermitian operator $\hat{B}$ in the same state

$$\sigma_B^2 = \langle (\hat{B} - \langle \hat{B} \rangle) \Psi | (\hat{B} - \langle \hat{B} \rangle) \Psi \rangle = \langle g | g \rangle$$

for $|g\rangle = |(\hat{B} - \langle \hat{B} \rangle) \Psi\rangle$.

The product of the two deviations can thus be expressed as

(1)

In order to relate the two vectors $|f\rangle$ and $|g\rangle$, we use the [Cauchy–Schwarz inequality](#)^[17] which is defined as

$$\langle f | f \rangle \langle g | g \rangle \geq |\langle f | g \rangle|^2,$$

and thus Equation (1) can be written as

(2)

Since $\langle f | g \rangle$ is in general a complex number, we use the fact that the modulus squared of any complex number z is defined as $|z|^2 = zz^*$, where z^* is the complex conjugate of z . The modulus squared can also be expressed as

(3)

we let $z = \langle f | g \rangle$ and $z^* = \langle g | f \rangle$ and substitute these into the equation above to get

(4)

The inner product $\langle f | g \rangle$ is written out explicitly as

$$\langle f | g \rangle = \langle (\hat{A} - \langle \hat{A} \rangle) \Psi | (\hat{B} - \langle \hat{B} \rangle) \Psi \rangle,$$

and using the fact that $\hat{A}$ and $\hat{B}$ are Hermitian operators, we find

$$\begin{aligned}
\langle f | g \rangle &= \langle \Psi | (\hat{A} - \langle \hat{A} \rangle)(\hat{B} - \langle \hat{B} \rangle) \Psi \rangle \\
&= \langle \Psi | (\hat{A}\hat{B} - \hat{A}\langle \hat{B} \rangle - \hat{B}\langle \hat{A} \rangle + \langle \hat{A} \rangle \langle \hat{B} \rangle) \Psi \rangle \\
&= \langle \Psi | \hat{A}\hat{B} \Psi \rangle - \langle \Psi | \hat{A} \langle \hat{B} \rangle \Psi \rangle - \langle \Psi | \hat{B} \langle \hat{A} \rangle \Psi \rangle + \langle \Psi | \langle \hat{A} \rangle \langle \hat{B} \rangle \Psi \rangle \\
&= \langle \hat{A}\hat{B} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle + \langle \hat{A} \rangle \langle \hat{B} \rangle \\
&= \langle \hat{A}\hat{B} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle.
\end{aligned}$$

Similarly it can be shown that $\langle g | f \rangle = \langle \hat{B}\hat{A} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle$.

Thus, we have

$$\langle f | g \rangle - \langle g | f \rangle = \langle \hat{A}\hat{B} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle - \langle \hat{B}\hat{A} \rangle + \langle \hat{A} \rangle \langle \hat{B} \rangle = \langle [\hat{A}, \hat{B}] \rangle$$

and

$$\langle f | g \rangle + \langle g | f \rangle = \langle \hat{A}\hat{B} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle + \langle \hat{B}\hat{A} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle = \langle \{\hat{A}, \hat{B}\} \rangle - 2\langle \hat{A} \rangle \langle \hat{B} \rangle$$

We now substitute the above two equations above back into Eq. (4) and get

$$|\langle f | g \rangle|^2 = \left(\frac{1}{2} \langle \{\hat{A}, \hat{B}\} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle \right)^2 + \left(\frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right)^2.$$

Substituting the above into Equation (2) we get the Schrödinger uncertainty relation

$$\sigma_A \sigma_B \geq \sqrt{\left(\frac{1}{2} \langle \{\hat{A}, \hat{B}\} \rangle - \langle \hat{A} \rangle \langle \hat{B} \rangle \right)^2 + \left(\frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right)^2}.$$

This proof has an issue^[18] related to the domains of the operators involved. For the proof to make sense, the vector $\hat{B}|\Psi\rangle$ has to be in the domain of the **unbounded operator** $\hat{A}$, which is not always the case. In fact, the Robertson uncertainty relation is false if $\hat{A}$ is an angle variable and $\hat{B}$ is the derivative with respect to this variable. In this example, the commutator is a nonzero constant—just as in the Heisenberg uncertainty relation—and yet there are states where the product of the uncertainties is zero.^[11] (See the counterexample section below.) This issue can be overcome by using a **variational method** for the proof,^{[19][20]} or by working with an exponentiated version of the canonical commutation relations.^[11]

Note that in the general form of the Robertson–Schrödinger uncertainty relation, there is no need to assume that the operators $\hat{A}$ and $\hat{B}$ are **self-adjoint operators**. It suffices

to assume that they are merely [symmetric operators](#). (The distinction between these two notions is generally glossed over in the physics literature, where the term *Hermitian* is used for either or both classes of operators. See Chapter 9 of Hall's book^[11] for a detailed discussion of this important but technical distinction.)

Phase space

In the [phase space formulation](#) of quantum mechanics, the Robertson–Schrödinger relation follows from a positivity condition on a real star-square function. Given a [Wigner function](#) $W(x, p)$ with [star product](#) $\star$ and a function f , the following is generally true:^[21]

$$\langle f^* \star f \rangle = \int (f^* \star f) W(x, p) dx dp \geq 0 .$$

Choosing $f = a + bx + cp$, we arrive at

$$\langle f^* \star f \rangle = \begin{bmatrix} a^* & b^* & c^* \end{bmatrix} \begin{bmatrix} 1 & \langle x \rangle & \langle p \rangle \\ \langle x \rangle & \langle x \star x \rangle & \langle x \star p \rangle \\ \langle p \rangle & \langle p \star x \rangle & \langle p \star p \rangle \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} \geq 0 .$$

Since this positivity condition is true for *all* a, b , and c , it follows that all the eigenvalues of the matrix are non-negative.

The non-negative eigenvalues then imply a corresponding non-negativity condition on the [determinant](#),

$$\det \begin{bmatrix} 1 & \langle x \rangle & \langle p \rangle \\ \langle x \rangle & \langle x \star x \rangle & \langle x \star p \rangle \\ \langle p \rangle & \langle p \star x \rangle & \langle p \star p \rangle \end{bmatrix} = \det \begin{bmatrix} 1 & \langle x \rangle & \langle p \rangle \\ \langle x \rangle & \langle x^2 \rangle & \left\langle xp + \frac{i\hbar}{2} \right\rangle \\ \langle p \rangle & \left\langle xp - \frac{i\hbar}{2} \right\rangle & \langle p^2 \rangle \end{bmatrix} \geq 0 ,$$

or, explicitly, after algebraic manipulation,

$$\sigma_x^2 \sigma_p^2 = (\langle x^2 \rangle - \langle x \rangle^2) (\langle p^2 \rangle - \langle p \rangle^2) \geq (\langle xp \rangle - \langle x \rangle \langle p \rangle)^2 + \frac{\hbar^2}{4} .$$

Examples

Since the Robertson and Schrödinger relations are for general operators, the relations can be applied to any two observables to obtain specific uncertainty relations. A few of the most common relations found in the literature are given below.

- **Position–linear momentum uncertainty relation:** for the position and linear momentum operators, the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$ implies the Kennard inequality from above:

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}.$$

- **Angular momentum uncertainty relation:** For two orthogonal components of the [total angular momentum](#) operator of an object:

$$\sigma_{J_i} \sigma_{J_j} \geq \frac{\hbar}{2} |\langle J_k \rangle|,$$

where i, j, k are distinct, and J_i denotes angular momentum along the x_i axis. This relation implies that unless all three components vanish together, only a single component of a system's angular momentum can be defined with arbitrary precision, normally the component parallel to an external (magnetic or electric) field. Moreover, for $[J_x, J_y] = i\hbar \epsilon_{xyz} J_z$, a choice $\hat{A} = J_x$, $\hat{B} = J_y$, in angular momentum multiplets, $\psi = |j, m\rangle$, bounds the [Casimir invariant](#) (angular momentum squared, $\langle J_x^2 + J_y^2 + J_z^2 \rangle$) from below and thus yields useful constraints such as $j(j+1) \geq m(m+1)$, and hence $j \geq m$, among others.

- For the number of electrons in a [superconductor](#) and the [phase](#) of its [Ginzburg–Landau order parameter](#)^{[22][23]}

$$\Delta N \Delta \varphi \geq 1.$$

Limitations

The derivation of the Robertson inequality for operators $\hat{A}$ and $\hat{B}$ requires $\hat{A}\hat{B}\psi$ and $\hat{B}\hat{A}\psi$ to be defined. There are quantum systems where these conditions are not valid.^[24] One example is a quantum [particle on a ring](#), where the wave function depends on an angular variable θ in the interval $[0, 2\pi]$. Define "position" and "momentum" operators $\hat{A}$ and $\hat{B}$ by

$$\hat{A}\psi(\theta) = \theta\psi(\theta), \quad \theta \in [0, 2\pi],$$

and

$$\hat{B}\psi = -i\hbar \frac{d\psi}{d\theta},$$

with periodic boundary conditions on $\hat{B}$. The definition of $\hat{A}$ depends the θ range from 0 to 2π . These operators satisfy the usual commutation relations for position and momentum operators, $[\hat{A}, \hat{B}] = i\hbar$. More precisely, $\hat{A}\hat{B}\psi - \hat{B}\hat{A}\psi = i\hbar\psi$ whenever both $\hat{A}\hat{B}\psi$ and $\hat{B}\hat{A}\psi$ are defined, and the space of such ψ is a dense subspace of the quantum Hilbert space.^[11]

Now let ψ be any of the eigenstates of $\hat{B}$, which are given by $\psi(\theta) = e^{2\pi i n \theta}$. These states are normalizable, unlike the eigenstates of the momentum operator on the line. Also the operator $\hat{A}$ is bounded, since θ ranges over a bounded interval. Thus, in the state ψ , the uncertainty of B is zero and the uncertainty of A is finite, so that

$$\sigma_A \sigma_B = 0.$$

The Robertson uncertainty principle does not apply in this case: ψ is not in the domain of the operator $\hat{B}\hat{A}$, since multiplication by θ disrupts the periodic boundary conditions imposed on $\hat{B}$.^[11]

For the usual position and momentum operators $\hat{X}$ and $\hat{P}$ on the real line, no such counterexamples can occur. As long as σ_x and σ_p are defined in the state ψ , the Heisenberg uncertainty principle holds, even if ψ fails to be in the domain of $\hat{X}\hat{P}$ or of $\hat{P}\hat{X}$.^[11]

Mixed states

The Robertson–Schrödinger uncertainty can be improved noting that it must hold for all components ϱ_k in any decomposition of the [density matrix](#) given as

$$\varrho = \sum_k p_k \varrho_k.$$

Here, for the probabilities $p_k \geq 0$ and $\sum_k p_k = 1$ hold. Then, using the relation

$$\sum_k a_k \sum_k b_k \geq \left(\sum_k \sqrt{a_k b_k} \right)^2$$

for $a_k, b_k \geq 0$, it follows that^[25]

$$\sigma_A^2 \sigma_B^2 \geq \left[\sum_k p_k L(\varrho_k) \right]^2,$$

where the function in the bound is defined

$$L(\varrho) = \sqrt{\left| \frac{1}{2} \operatorname{tr}(\rho\{A, B\}) - \operatorname{tr}(\rho A) \operatorname{tr}(\rho B) \right|^2 + \left| \frac{1}{2i} \operatorname{tr}(\rho[A, B]) \right|^2}.$$

The above relation very often has a bound larger than that of the original Robertson–Schrödinger uncertainty relation. Thus, we need to calculate the bound of the Robertson–Schrödinger uncertainty for the mixed components of the quantum state rather than for the quantum state, and compute an average of their square roots. The following expression is stronger than the Robertson–Schrödinger uncertainty relation

$$\sigma_A^2 \sigma_B^2 \geq \left[\max_{p_k, \varrho_k} \sum_k p_k L(\varrho_k) \right]^2,$$

where on the right-hand side there is a concave roof over the decompositions of the density matrix. The improved relation above is saturated by all single-qubit quantum states.^[25]

With similar arguments, one can derive a relation with a convex roof on the right-hand side^[25]

$$\sigma_A^2 F_Q[\varrho, B] \geq 4 \left[\min_{p_k, \Psi_k} \sum_k p_k L(|\Psi_k\rangle\langle\Psi_k|) \right]^2$$

where $F_Q[\varrho, B]$ denotes the [quantum Fisher information](#) and the density matrix is decomposed to pure states as

$$\varrho = \sum_k p_k |\Psi_k\rangle\langle\Psi_k|.$$

The derivation takes advantage of the fact that the quantum Fisher information is the convex roof of the variance times four.^{[26][27]}

A simpler inequality follows without a convex roof^[28]

$$\sigma_A^2 F_Q[\varrho, B] \geq |\langle i[A, B] \rangle|^2,$$

which is stronger than the Heisenberg uncertainty relation, since for the quantum Fisher information we have

$$F_Q[\varrho, B] \leq 4\sigma_B^2,$$

while for pure states the equality holds.

The Maccone–Pati uncertainty relations

The Robertson–Schrödinger uncertainty relation can be trivial if the state of the system is chosen to be eigenstate of one of the observable. The stronger uncertainty relations proved by Lorenzo Maccone and [Arun K. Pati](#) give non-trivial bounds on the sum of the variances for two incompatible observables.^[29] (Earlier works on uncertainty relations formulated as the sum of variances include, e.g., Ref.^[30] due to Yichen Huang.) For two non-commuting observables $\mathbf{A}$ and $\mathbf{B}$ the first stronger uncertainty relation is given by

$$\sigma_A^2 + \sigma_B^2 \geq \pm i \langle \Psi | [\mathbf{A}, \mathbf{B}] | \Psi \rangle + |\langle \Psi | (\mathbf{A} \pm i\mathbf{B}) | \bar{\Psi} \rangle|^2,$$

where $\sigma_A^2 = \langle \Psi | \mathbf{A}^2 | \Psi \rangle - \langle \Psi | \mathbf{A} | \Psi \rangle^2$, $\sigma_B^2 = \langle \Psi | \mathbf{B}^2 | \Psi \rangle - \langle \Psi | \mathbf{B} | \Psi \rangle^2$, $|\bar{\Psi}\rangle$ is a normalized vector that is orthogonal to the state of the system $|\Psi\rangle$ and one should choose the sign

of $\pm i\langle \Psi | [A, B] | \Psi \rangle$ to make this real quantity a positive number.

The second stronger uncertainty relation is given by

$$\sigma_A^2 + \sigma_B^2 \geq \frac{1}{2} |\langle \bar{\Psi}_{A+B} | (A + B) | \Psi \rangle|^2$$

where $|\bar{\Psi}_{A+B}\rangle$ is a state orthogonal to $|\Psi\rangle$. The form of $|\bar{\Psi}_{A+B}\rangle$ implies that the right-hand side of the new uncertainty relation is nonzero unless $|\Psi\rangle$ is an eigenstate of $(A + B)$. One may note that $|\Psi\rangle$ can be an eigenstate of $(A + B)$ without being an eigenstate of either A or B . However, when $|\Psi\rangle$ is an eigenstate of one of the two observables the Heisenberg–Schrödinger uncertainty relation becomes trivial. But the lower bound in the new relation is nonzero unless $|\Psi\rangle$ is an eigenstate of both.

Energy–time

An energy–time uncertainty relation like

$$\Delta E \Delta t \gtrsim \hbar/2,$$

has a long, controversial history; the meaning of Δt and ΔE varies and different formulations have different arenas of validity.^[31] However, one well-known application is both well established^{[32][33]} and experimentally verified:^{[34][35]} the connection between the life-time of a resonance state, $\tau_{\sqrt{1/2}}$ and its energy width ΔE :

$$\tau_{\sqrt{1/2}} \Delta E = \pi \hbar/4.$$

In particle-physics, widths from experimental fits to the [Breit–Wigner energy distribution](#) are used to characterize the lifetime of quasi-stable or decaying states.^[36]

An informal, heuristic meaning of the principle is the following:^[37] A state that only exists for a short time cannot have a definite energy. To have a definite energy, the frequency of the state must be defined accurately, and this requires the state to hang around for many cycles, the reciprocal of the required accuracy. For example, in [spectroscopy](#), excited states have a finite lifetime. By the time–energy uncertainty principle, they do not have a definite energy, and, each time they decay, the energy they release is slightly different. The average energy of the outgoing photon has a peak at the theoretical energy of the state, but the distribution has a finite width called the [natural linewidth](#). Fast-decaying states have a broad linewidth, while slow-decaying states have a narrow linewidth.^[38] The same linewidth effect also makes it difficult to specify the [rest mass](#) of unstable, fast-decaying particles in [particle physics](#). The faster the [particle decays](#) (the shorter its lifetime), the less certain is its mass (the larger the particle's [width](#)).

Time in quantum mechanics

The concept of "time" in quantum mechanics offers many challenges.^[39] There is no quantum theory of time measurement; relativity is both fundamental to time and difficult to include in quantum mechanics.^[31] While position and momentum are associated with a single particle, time is a system property: it has no operator needed for the Robertson–Schrödinger relation.^[1] The mathematical treatment of stable and unstable quantum systems differ.^[40] These factors combine to make energy–time uncertainty principles controversial.

Three notions of "time" can be distinguished:^[31] external, intrinsic, and observable. External or laboratory time is seen by the experimenter; intrinsic time is inferred by changes in dynamic variables, like the hands of a clock or the motion of a [free particle](#); observable time concerns time as an observable, the measurement of time-separated events.

An external-time energy–time uncertainty principle might say that measuring the energy of a quantum system to an accuracy ΔE requires a time interval $\Delta t > \hbar/\Delta E$.^[33] However, [Yakir Aharonov](#) and [David Bohm](#)^{[41][31]} have shown that, in some quantum systems, energy can be measured accurately within an arbitrarily short time: external-time uncertainty principles are not universal.

Intrinsic time is the basis for several formulations of energy–time uncertainty relations, including the Mandelstam–Tamm relation discussed in the next section. A physical system with an intrinsic time closely matching the external laboratory time is called a "clock".^{[39]:31}

Observable time, measuring time between two events, remains a challenge for quantum theories; some progress has been made using [positive operator-valued measure](#) concepts.^[31]

Mandelstam–Tamm

In 1945, [Leonid Mandelstam](#) and [Igor Tamm](#) derived a non-relativistic *time–energy uncertainty relation* as follows.^{[42][31]} From Heisenberg mechanics, the generalized [Ehrenfest theorem](#) for an observable B without explicit time dependence, represented by a self-adjoint operator $\hat{B}$ relates time dependence of the average value of $\hat{B}$ to the average of its commutator with the Hamiltonian:

$$\frac{d\langle \hat{B} \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{B}] \rangle.$$

The value of $\langle [\hat{H}, \hat{B}] \rangle$ is then substituted in the [Robertson uncertainty relation](#) for the energy operator $\hat{H}$ and $\hat{B}$:

$$\sigma_H \sigma_B \geq \left| \frac{1}{2i} \langle [\hat{H}, \hat{B}] \rangle \right|,$$

giving

$$\sigma_H \frac{\sigma_B}{\left| \frac{d\langle \hat{B} \rangle}{dt} \right|} \geq \frac{\hbar}{2}$$

(whenever the denominator is nonzero). While this is a universal result, it depends upon the observable chosen and that the deviations σ_H and σ_B are computed for a particular state. Identifying $\Delta E \equiv \sigma_E$ and the characteristic time

$$\tau_B \equiv \frac{\sigma_B}{\left| \frac{d\langle \hat{B} \rangle}{dt} \right|}$$

gives an energy–time relationship $\Delta E \tau_B \geq \frac{\hbar}{2}$. Although τ_B has the dimension of time, it is different from the time parameter t that enters the [Schrödinger equation](#). This τ_B can be interpreted as time for which the expectation value of the observable, $\langle \hat{B} \rangle$, changes by an amount equal to one standard deviation.^[43] Examples:

- The time a free quantum particle passes a point in space is more uncertain as the energy of the state is more precisely controlled: $\Delta T = \hbar/2\Delta E$. Since the time spread is related to the particle position spread and the energy spread is related to the momentum spread, this relation is directly related to position–momentum uncertainty.^{[16]:144}
- A [Delta particle](#), a quasistable composite of quarks related to protons and neutrons, has a lifetime of 10^{-23} s, so its measured [mass equivalent to energy](#), 1232 MeV/c², varies by ± 120 MeV/c²; this variation is intrinsic and not caused by measurement errors.^{[16]:144}
- Two energy states $\psi_{1,2}$ with energies $E_{1,2}$, superimposed to create a composite state

$$\Psi(x, t) = a\psi_1(x)e^{-iE_1t/\hbar} + b\psi_2(x)e^{-iE_2t/\hbar}.$$

The [probability amplitude](#) of this state has a time-dependent interference term:

$$|\Psi(x, t)|^2 = a^2 |\psi_1(x)|^2 + b^2 |\psi_2(x)|^2 + 2ab \cos\left(\frac{E_2 - E_1}{\hbar} t\right).$$

The oscillation period varies inversely with the energy difference: $\tau = 2\pi\hbar/(E_2 - E_1)$.^{[16]:144}

Each example has a different meaning for the time uncertainty, according to the observable and state used.

Quantum field theory

Some formulations of [quantum field theory](#) uses temporary electron–positron pairs in its calculations called [virtual particles](#). The mass-energy and lifetime of these particles are related by the energy–time uncertainty relation. The energy of a quantum systems is not known with enough precision to limit their behavior to a single, simple history. Thus the influence of *all histories* must be incorporated into quantum calculations, including those with much greater or much less energy than the mean of the measured/calculated energy distribution.

The energy–time uncertainty principle does not temporarily violate [conservation of energy](#); it does not imply that energy can be "borrowed" from the universe as long as it is "returned" within a short amount of time.^{[16]:145} The energy of the universe is not an exactly known parameter at all times.^[1] When events transpire at very short time intervals, there is uncertainty in the energy of these events.

Harmonic analysis

In the context of [harmonic analysis](#) the uncertainty principle implies that one cannot at the same time localize the value of a function and its Fourier transform. To wit, the following inequality holds,

$$\left(\int_{-\infty}^{\infty} x^2 |f(x)|^2 dx \right) \left(\int_{-\infty}^{\infty} \xi^2 |\hat{f}(\xi)|^2 d\xi \right) \geq \frac{\|f\|_2^4}{16\pi^2}.$$

Further mathematical uncertainty inequalities, including the above [entropic uncertainty](#), hold between a function f and its Fourier transform $\hat{f}$:^{[44][45][46]}

$$H_x + H_\xi \geq \log(e/2)$$

Signal processing

In the context of [time–frequency analysis](#) uncertainty principles are referred to as the **Gabor limit**, after [Dennis Gabor](#), or sometimes the *Heisenberg–Gabor limit*. The basic result, which follows from "Benedicks's theorem", below, is that a function cannot be both [time limited](#) and [band limited](#) (a function and its Fourier transform cannot both have bounded domain)—see [bandlimited versus timelimited](#). More accurately, the *time-bandwidth* or *duration-bandwidth* product satisfies

$$\sigma_t \sigma_f \geq \frac{1}{4\pi} \approx 0.08 \text{ cycles},$$

where σ_t and σ_f are the standard deviations of the time and frequency energy concentrations respectively.^[47] The minimum is attained for a [Gaussian](#)-shaped pulse ([Gabor wavelet](#)) [For the un-squared Gaussian (i.e. signal amplitude) and its un-squared Fourier transform magnitude